Articulating the Role of Artificial Intelligence in Collective Intelligence: A Transactive Systems Framework

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Human society faces increasingly complex problems that require coordinated collective action. Artificial intelligence (AI) holds the potential to bring together the knowledge and associated action needed to find solutions at scale. In order to unleash the potential of human and AI systems, we need to understand the core functions of collective intelligence. To this end, we describe a socio-cognitive architecture that conceptualizes how boundedly rational individuals coordinate their cognitive resources and diverse goals to accomplish joint action. Our transactive systems framework articulates the inter-member processes underlying the emergence of collective memory, attention, and reasoning, which are fundamental to intelligence in any system. Much like the cognitive architectures that have guided the development of artificial intelligence, our transactive systems framework holds the potential to be formalized in computational terms to deepen our understanding of collective intelligence and pinpoint roles that AI can play in enhancing it.

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INTRODUCTION

Over the last several decades, the use of groups, teams, and larger collectives has grown rapidly in every sector of society, to deliver emergency services, defend against physical and cyber-attacks, detect and vaccinate against major diseases, or develop scientific knowledge (National Research Council, 2015). With the increasing dynamism of our globally-interconnected environment, these groups constantly face new emergencies or threats requiring new forms of large-scale, coordinated response. Consequently, established organizational structures and pre-scripted routines are set aside as new collectives form and improvise more flexible ways to organize, often across or outside of organizational boundaries (Marks et al., 2005).

Reflecting this reality, management theory is coming to recognize that groups and organizations operate less like a "machine" and more like a complex adaptive system (Mathieu et al., 2019). Consequently, we need to shift from designing structures that are optimized for narrowly defined performance, and instead create self-regulating, adaptive systems designed for collective intelligence. Conceptualizing the ways that systems can be designed for collective intelligence helps to identify roles that artificial intelligence can play in enhancing their functioning.

Intelligence, whether in the context of an individual, a collective, or a technological system, is broadly defined as the ability to perform or achieve goals in a wide range of environments (Legg & Hutter, 2007). Implicit in "ability to perform" is the notion of adapting in order to maintain performance, and "wide variety of environments" refers to tasks and environments of varying complexity. Thus, high intelligence is signaled by sustained performance in the face of changes in complexity over time.

Three functions that any intelligent system needs to manage are memory, attention, and reasoning. These are core functions that characterize the systems of the human brain (Luria, 1973) but they are essential in other intelligent systems as well, whether biological, technological (AI), or human-AI (Malone & Bernstein, 2015). In human systems, individuals can augment their limited cognitive memory, attention and reasoning abilities with metacognitive abilities that enable them to engage in interdependent transactive processes with others. We theorize that collective intelligence is fostered by the emergence and mutual adaptation of these transactive memory, attention, and reasoning systems. These processes allow collectives to dynamically adapt to the complexity of their environment by efficiently coordinating their available resources to collectively pursue rewarding goals while fulfilling member needs.

Furthermore, with increasing developments in artificial intelligence (AI), there has been enormous growth in the use of autonomous and semi-autonomous systems interacting with humans on complex interdependent tasks. In current applications, these typically take the form of human-autonomy teams (McNeese et al., 2021) in which synthetic team members play a well-defined role in carrying out taskwork. However, we posit that in addition to carrying out taskwork, AI offers the potential for helping to facilitate *teamwork*, particularly in dynamic environments that lack the well-defined organizational structures traditionally relied upon to enhance humans' bounded rationality (Simon, 1957). The capability to facilitate teamwork would be greatly enhanced by the development of artificial social intelligence (ASI) which enables inferences about human mental states, including goals, situational knowledge, beliefs and emotions of human partners. We contend that a transactive systems framework for collective intelligence enables us to identify the unique role that ASI can play in enhancing teamwork, which will in turn greatly expand the capacity for collective intelligence in the complex, dynamic environments where it is so desperately needed.

In this paper, we briefly review and integrate the relevant literature to describe our novel transactive systems framework for the emergence of collective memory, attention, and reasoning underlying collective intelligence. We then build on the framework to describe the capacities in which artificial intelligence—particularly ASI—can augment these systems to increase collective intelligence.

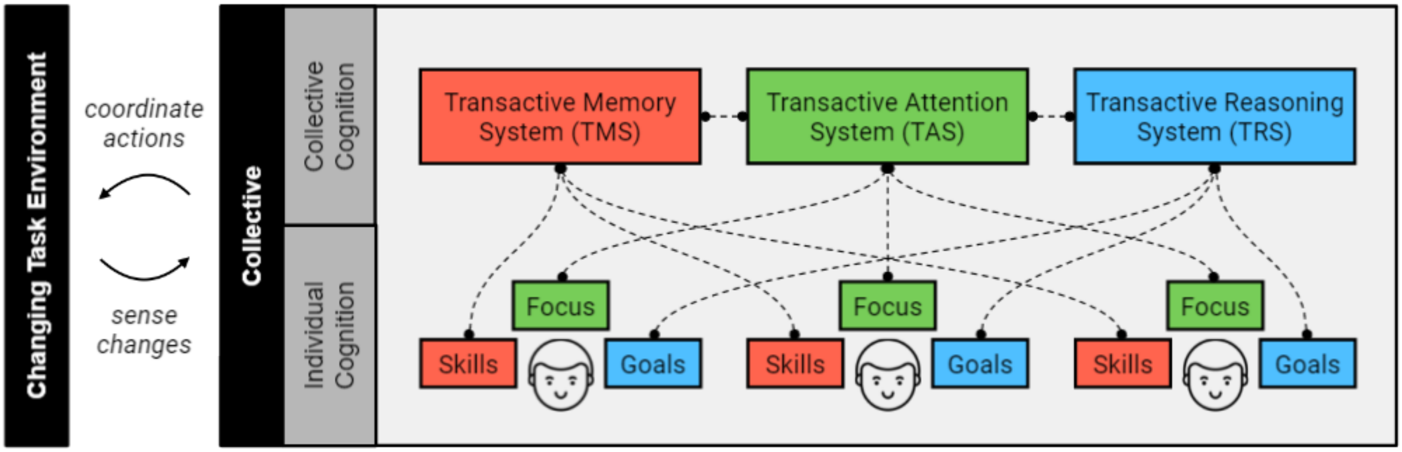

Figure 1: Transactive Systems Framework for Collective intelligence

TRANSACTIVE SYSTEMS FRAMEWORK FOR COLLECTIVE INTELLIGENCE

Collective intelligence has been conceptualized and measured in a variety of different contexts, including organizations (Knott, 2008; Mayo & Woolley, 2021), networks (Kabo, 2018; Pescetelli et al., 2021), crowds (Kittur et al., 2009), and small groups (Riedl et al., 2021; Woolley et al., 2010). The design problem of collective intelligence in all of these settings is to put together a set of heterogeneous members who can coordinate their distributed cognitive resources (i.e., memory and attention) and diverse goals to formulate a series of joint actions that fulfill its efficiency and maintenance functions over time. While not always explicitly conceptualized using the same terminology, we contend that these functional processes underlie the emergence of CI in all of the contexts we've mentioned, and that an organizing framework will enable us to more deliberately enhance CI using emergent AI capabilities. Below we briefly describe the three transactive systems that handle the collective memory, attention, and reasoning functions that underlie the emergence of collective intelligence (see Figure 1). Each of these systems emerges via transactive processes that drive the updating, allocation, and retrieval of the related cognitive resource; we will describe how these manifest in each socio-cognitive function below.

Transactive memory system (TMS). TMS is a form of individually-held cognitive understanding of team members' skills, whereby each member has access to who knows what on the team and can retrieve that knowledge to achieve effective coordination (Ren & Argote, 2011). TMS is a dynamic system consisting of the directory of each members' knowledge and skills relevant to the task and the inter-member processes that facilitate allocation and retrieval of information to and from the most appropriate member. While the concept was initially developed in the context of couples in close relationships, it has since been extended to the group level (Ren & Argote, 2011) and consistently predicts team performance.

The TMS involves three transactive processes that ensure effective utilization of members' limited and distributed knowledge and skills. (1) Updating 'who knows what' occurs as members work together on interdependent tasks and information about everyones' competencies is observed and stored. (2) Allocation involves routing new, incoming information to the member best positioned to act on it. The typical consequence of directing and accumulating closely related knowledge within the same person is increased specialization. (3) Retrieval is aided by well-functioning updating and allocation, which minimizes the time required for retrieving knowledge. With multiple retrieval episodes, members come to form shared beliefs about the extent to which they can credibly rely on a given member's expertise. These features (specialization and credibility) are often used as behavioral indicators of a well-developed TMS (Ren & Argote, 2011).

Transactive Attention System (TAS). In addition to coordinating skills, groups also need to allocate member attention to accomplish work. Attention management is argued to be central to the role of management in any social system and is at the heart of organizational design decisions (Simon, 1973). In any information processing system, the available attention of members puts a hard limit on the capacity of the system to handle information and conduct work. Thus, TAS is a dynamic system consisting of members' attentional capacity, the signaling of different members' focus and availability, and the inter-member, transactive processes that facilitate allocation of individual attention and retrieval of joint attention when needed. TAS is theorized as a complement to TMS as groups distribute tasks and information, balancing between optimizing use of member expertise and attentional resources.

Just as with TMS, TAS involves three inter-member transactive processes that facilitate the efficient utilization of collective attention. (1) Updating understanding of member availability involves developing systems to keep all informed of each members' ongoing workload. (2) Attention allocation requires a consistent and shared system or set of norms to signal and update attentional priorities (which are informed by the TRS; see next section). This enables the collective to identify gaps and bottlenecks. As the number of members in the collective increases, these systems become increasingly important, as the amount of availability and allocation information to process increases exponentially. (3) Attention retrieval is necessary for tasks that require synchronous work, and is aided by an effective updating system. Groups with a strong attention retrieval capability will exhibit organized patterns of synchronous attention, sometimes manifest as "burstiness" where periods of independent work are punctuated by periods where members are simultaneously active on shared work and highly responsive to each others' requests. In recent work, burstiness has been shown to predict CI (Mayo & Woolley, 2021; Riedl & Woolley, 2017).

Transactive Reasoning System (TRS). While TMS and TAS work together to ensure efficient and effective utilization of the available member resources, they do not guarantee that the team is pursuing the right goals or that its members are highly motivated to go after these goals. Tackling internal and external uncertainty is foundational to a system's ability to maintain itself. Collective reasoning serves these functions: it evaluates collective goals in the context of a constantly changing environment to ensure the pursuit of those with the greatest value, and it facilitates alignment between individual and collective goals (Bacharach, 1999; Locke & Latham, 1990). Thus, we propose TRS as a dynamic system consisting of member's knowledge of their own and others' goals and motivational needs, and transactive processes whereby members maximize joint rewards via negotiation and alignment around these goals and priorities.

The TRS involves three inter-member transactive processes that facilitate collective reasoning. (1) Updating within TRS occurs through observations of the environment and observations of other members. Research at the intersection of motivation and entrepreneurship suggests that highly motivated individuals are better at recognizing and creating opportunities from their environment (A. Carsrud et al., 2009). In addition, through experience and interaction with other members, individuals will observe what others are more responsive to or learn what they care about and draw inferences about their goals. (2) Allocation with respect to TRS occurs via the negotiation and prioritization of goals to ensure focus on those that are most rewarding to the collective as a whole. Such goal clarity would also allow them to better recognize the missing resources needed to achieve their joint goals. (3) Retrieval with regard to TRS comes in the form of commitment to the collective and its objectives. Recent work on transactive goal dynamics has articulated how individuals in interdependent relationships form a single unit wherein they adopt and hold other- and system-oriented goals (Fitzsimons et al., 2015). When members adopt a dense set of goals that are aligned with others' goals, the collective will be able to devote more resources towards their joint goals and is more likely to experience successful outcomes. This in turn increases commitment to the collective. Ultimately, the guidance on goals and priorities from the TRS shapes decisions about resources to enable the TAS and TMS processes to effectively execute.

ARTIFICIAL INTELLIGENCE AND CI

The transactive systems framework for collective intelligence holds the potential to be formalized and further developed to deepen our understanding of CI. Though much of the existing work we draw upon is informed primarily by observation of human systems, we believe the socio-cognitive architecture we have articulated suggests the functions AI could fulfill to enhance collective intelligence.

The role of artificial intelligence is growing in almost every area of society. In thinking about the different classes of artificial intelligence technologies that have been developed, one major distinction often made is between "production" technologies and "coordination" technologies (Scott et al., 2007). Production technologies are those that directly help carry out taskwork, either by taking over tasks for humans altogether or by making the humans more accurate or efficient in completing tasks. Most technological development begins here, whether we think of early generations of technology (hand tools, machines) or those of the digital age. By contrast, coordination technologies are those that enable the connection or coordination of effort or information from different sources. Some current examples of coordination technologies include algorithms that match people seeking a good or service with the best source to provide it (such as ridesharing or online market platforms). Such technologies offer the best opportunities for enhancing teamwork. While few have been developed in this direction, we see additional roles that AI can play in enhancing socio-cognitive processes and CI, resulting in better teamwork (Fiore & Wiltshire, 2016).

We articulate three different types of roles we see for AI in enhancing collective intelligence—as an Assistive AI that augments individuals' production capabilities, as a Coach AI that nudges team collaboration behaviors, and as a Diagnostic AI that monitors indicators of collaborative process. We describe each by drawing on examples from recent literature as well as describing future possibilities, guided by the demands of the transactive systems underlying CI.

Assistive AI. People are increasingly using what we refer to in Figure 2 as "Assistive AI." These are used to augment all of the cognitive functions mentioned in individual task performance, such as internet search algorithms to aid memory and find information, reminders to direct attention, or even tools that track information in the environment (e.g. financial

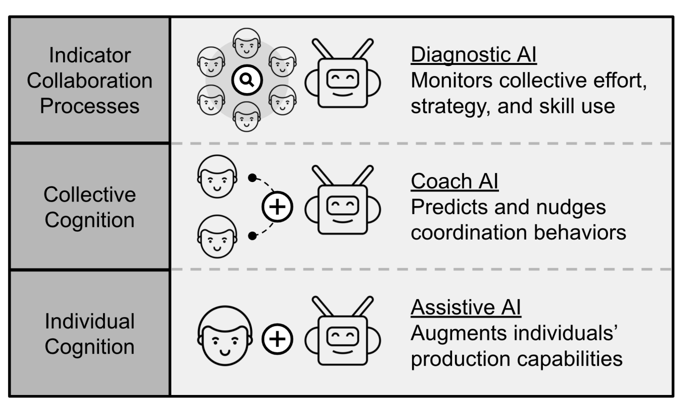

Figure 2: Potential roles of artificial intelligence (AI) agents for enhancing collective intelligence.

indicators, job postings) that inform individual goals. We could also foresee assistive AI that might incorporate knowledge of individual characteristics or preferences to specifically augment certain areas of weakness. For instance, an assistive AI that knows about an individual's goal to finish a difficult writing project could provide encouragement in the form of feedback about progress, reminders, and perhaps highlight activities or information that the user enjoys as a reward for accomplishing segments of the work.

Coach AI. Another role for technology is a "Coach AI" that could help enhance transactive memory, attention, and reasoning in teams. With respect to transactive memory, some new tools enhance this by sensing the content of one member's work and proactively alerting them to related information team members might have (e.g., Gimpel et al., 2020). An important calibration in such systems is to supply the right amount of information, as supplying too many connections could easily lead to overload and defeat the intent of enhanced CI. Tools that foster transactive attention include those supporting "flash teams" (Retelny et al., 2014) which break complex, multi-faceted projects into segments and coordinate task assignments and hand-offs among contributors. These tools have made it possible to accomplish fairly complex projects in significantly less time than standard. Finally, in the area of transactive reasoning, some AI-based "coach" tools help teams identify what different members prefer, for instance by using market-like bidding systems (Malone et al., 2017) to identify a goal members value most highly.

Another approach to developing Coach AI would be to proactively shape behavior; the best coaches structure or initiate action proactively rather than waiting for difficulties to develop (Hackman & Wageman, 2005). Following this, some researchers have experimented with relatively simple "nudges" to proactively shape norms or prime awareness of collective memory, attention, or reasoning as a means of sparking the associated transactive processes (Gupta et al., 2019). Further advances of Coach AI will be possible with additional development of ASI. An ASI capability will enable other tools to better sense user mental states and anticipate what problems collaborators might face. For instance, a tool that is able to sense dramatic changes in emotional state, such as anger, might trigger the transactive reasoning or transactive attention processes to determine if collective goals need to be revisited or task allocations realigned.

Diagnostic AI. Beyond shaping behavior, there are also possibilities for developing or leveraging AI to monitor a team or larger group via observable collaborative process indicators to diagnose early signs of trouble. Such diagnostics could in turn trigger interventions via Coach AI. Recent research has identified three collaborative processes that are good diagnostic indicators of collective functioning and consistently associated with CI, including the *level of collective effort*, coordination of *appropriate task strategy,* and matching members to tasks and roles to achieve *appropriate skill use* (Hackman & Wageman, 2005; Riedl et al., 2021). These processes have been identified as the major sources of "process loss" in a wide range of collaborative settings and thus can be key leverage points for intervention. For example, monitoring the level of effort and motivation by tracking member activity levels or emotional states (Van Kleef et al., 2012) can provide early indications of diminishing motivation levels and trigger a review of goal clarity and alignment. Indicators of coordination via activity patterns (Mayo & Woolley, 2021) can be diagnostic of the appropriateness of task strategy and prompt a review of systems for managing attention. Finally, the match between member expertise and time spent on tasks requiring that expertise, along with an increase in specialization among group members, can help diagnose appropriate skill use. Deficits in this can prompt the initiation of corrective steps such as team training or the provision of additional information systems to help manage and direct information encoding, storage, and retrieval (Ren & Argote, 2011). Furthermore, with additional development of Diagnostic AI capabilities, such a system could take in many indicators such as those described and decide if, when, and how to intervene.

With the continued development of AI, it is certainly possible to imagine systems that could completely "manage" and organize the interactions of group members to routinely create high levels of CI. However, we also know humans seek autonomy and need to trust technological systems before complying with them. For instance, work on algorithm aversion demonstrates the numerous situations in which workers override tools and technologies even when doing so to their detriment (Glikson & Woolley, 2020)**.** Systems that would deprive workers of their autonomy to exercise judgment or experience responsibility for their results can undermine their sense of internal motivation (Hackman & Oldham, 1976)**.** Thus it would seem that the introduction of AI-driven coaching or diagnostic systems will need to strike the right balance between autonomy and control, as well as not undermine the natural social cognition normally developed through collaboration (Gupta & Woolley, 2018)**.**

In conclusion, much like the cognitive architectures that have guided the development of artificial intelligence, our transactive systems framework holds the potential to be formalized in computational terms to deepen our understanding of collective intelligence. Such a socio-cognitive architecture not only suggests the functions AI could fulfill but also suggests new directions for enhancing human-AI systems.

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